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Mapping Losses through Empirical Extraction of the Spatial External Luminescence Efficiency

Tamir Yeshurun, Mor Fiegenbaum-Raz, and Gideon Segev*

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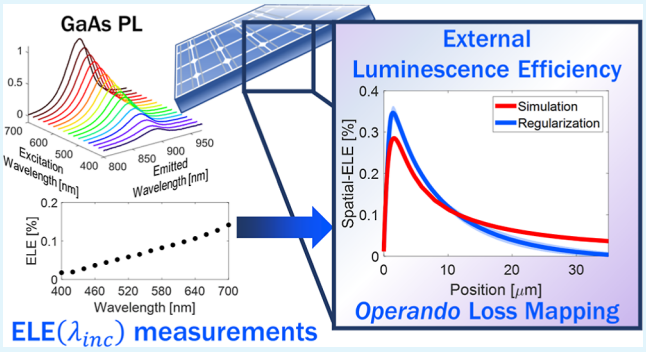
ABSTRACT: The performance of photovoltaic cells is determined by the loss mechanisms and charge transport properties within the device. Thus, quantitative assessment of these mechanisms is fundamental for device optimization and the evaluation of new materials and interfaces. This paper introduces a method for the in-depth quantitative mapping of loss and charge transport mechanisms under operando conditions. The spatial external luminescence efficiency (SELE) is defined as the probability for an electron−hole pair that is photogenerated at a specific depth within a device to contribute to its photoluminescence. By coupling incident wavelength-dependent external luminescence efficiency measurements with optical modeling, we demonstrate the extraction of the SELE of a GaAs wafer. The SELE directly maps the competition between radiative and nonradiative recombination within the device and can thus be used to study the properties of surfaces and interfaces. Moreover, since the external luminescence efficiency is related to the obtainable photovoltage from the device, the SELE can be used to map the spatial contribution to the device photovoltage. Furthermore, the SELE is calculated by using device simulations that account for photon recycling. These simulations validate the extraction method and attribute the features in the profile to important physical quantities, such as the surface recombination velocity and carrier lifetime.



KEYWORDS: *operando characterization, luminescence efficiency, internal optical losses, loss mapping, photovoltage*

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1. INTRODUCTION

The search for new light absorbers and selective contacts is key to the development of next-generation, high-efficiency solar cells.[1] The performance of these devices is determined by the competition between charge carrier collection and various recombination processes throughout the device. Thus, in-depth and rapid characterization of the optoelectronic properties of materials and interfaces can accelerate this development and identify performance bottlenecks. Previous works have shown that a careful examination of the spatial collection efficiency (SCE) can be used to map losses in photovoltaic (PV) and photoelectrochemical (PEC) cells.[2−9] The SCE is defined as the fraction of charge carriers that are optically generated at a given location and contribute to current generation. Thus, the SCE provides a functional depth profile of the active regions in the device under operando conditions. Analysis of SCE profiles allowed estimating diffusion and collection lengths, differentiating between bulk and surface losses, and quantifying the probability that absorbed photons do not generate mobile charge carriers in metal oxides.[8−12] Although the SCE provides extensive information on current collection, it cannot be used to characterize surface passivation layers, which are essential for reducing surface recombination losses, nor to provide information on the contribution to the cell voltage. Hence, a new complementary characterization method that can map and quantify these properties is called for.

Under standard operating conditions, photogenerated electron−hole pairs that are not collected as current recombine through bulk and surface defect states or through radiative recombination, thus generating photons. These photons are either reabsorbed within the device (namely, photon recycling) or emitted through photoluminescence (PL). The fraction of the electron hole pairs that recombine radiatively is the internal luminescence yield, η_{int}, and the probability for any recombination event to result in photon emission is the external luminescence efficiency, η_{ext} (ELE).[13−16] Since both the radiative recombination rate and the photovoltage are determined by the separation of electron and hole quasi-Fermi

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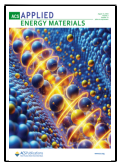

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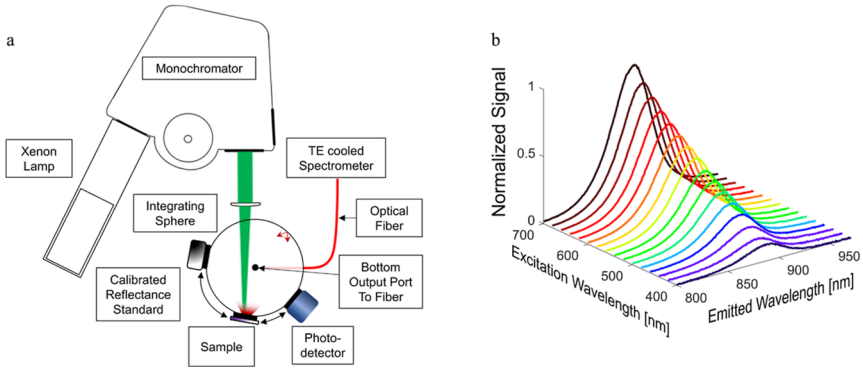

Figure 1. (a) Incident wavelength-dependent PL measurement setup using an integrating sphere. The photodetector and reflectance standard are used for incident photon-flux measurements and calibration. (b) PL measurements of a p-type GaAs sample for different excitation wavelengths.

levels, the open-circuit voltage that can be obtained from a light absorber, V_{oc}, is tightly coupled to its ELE[15,17]

$$qV_{oc} = qV_{oc,ideal} + kT\ln(\eta_{ext}) \tag{1}$$

where $V_{oc,ideal}$ is the open-circuit voltage of an ideal light absorber, k is the Boltzmann constant, q is the fundamental charge, and T is the device temperature. Thus, ELE measurements provide an estimate of the maximum open-circuit voltage that can be obtained from a light-absorbing material.[18] Consequently, device optimization toward maximizing the extraction of radiative recombination photons led to a significant increase in the performance of GaAs solar cells, achieving a record efficiency of 28.8%.[16]

Various approaches were developed to study important device parameters by using PL measurements. Voltage-dependent PL imaging was used to expose nonideal charge extraction regions.[19,20] A lateral map of the depth-averaged quasi-Fermi-level separation was obtained using hyper-spectral PL imaging.[21] Lombez et al. showed that combining voltage-dependent luminescence measurements with electrical transport simulations can provide a qualitative depth analysis of device properties.[22] Tennyson et al. produced tomographic photovoltage images of CIGS solar cells by acquiring photovoltage maps of devices illuminated at different wavelengths.[23] However, since there was no attempt to deconvolute the correlation between the optical generation depth profile and the obtained photovoltage, only qualitative depth information was obtained. Despite providing a wealth of information about the device performance and material properties, these methods do not map losses quantitatively along the depth profile of the device.

The spatial external luminescence efficiency (SELE) is a depth profile of the contribution to the ELE. In this work, we show how the SELE can be extracted by combining incident wavelength-dependent ELE measurements with optical modeling. The SELE of a GaAs wafer, which serves as a model system, is extracted and analyzed. We show that the SELE profiles quantify the competition between radiative and nonradiative recombination processes at every point within the device, as well as the spatial contribution to the photovoltage. Finite-element simulations are used to verify the extracted profile and relate features in the SELE profile to different material properties such as diffusion length and surface recombination velocity.

1.1. Spatial External Luminescence Efficiency. The SELE, $S(z)$, is defined as the probability for photogenerated electron hole pairs that are generated at some point z within a device to contribute to the measured PL photon flux, ϕ_{out}. Therefore, the total PL photon flux induced by illumination with a monochromatic incident beam at wavelength λ_{in} is as follows

$$\phi_{out}(\lambda_{in}) = \int_0^d \Delta G(\lambda_{in}, z)S(z)\mathrm{d}z \tag{2}$$

where $\Delta G(\lambda_{in}, z)$ is the optical generation that is induced by the monochromatic light, and d is the device thickness. The SELE is the correlation between the photoexcitation position in which charge transport processes commence and the eventual contribution to photoemission after undergoing transport processes. Thus, SELE profiles account for the effects of drift and diffusion, all recombination mechanisms, surfaces, and junction formations. Furthermore, SELE profiles extracted from eq 2 account for these mechanisms with no assumptions on the charge transport mechanisms, position of junctions, or the effects of surfaces.

Under open-circuit conditions, ELE is the fraction of absorbed photons that are emitted in the PL process

$$\eta_{ext}(\lambda_{in}) = \frac{\phi_{out}(\lambda_{in})}{\phi_{abs}(\lambda_{in})} \tag{3}$$

$\phi_{abs} = \phi_0(\lambda_{in})A(\lambda_{in})$ is the flux of photons that are absorbed in the sample, where $\phi_0(\lambda_{in})$ is the incident monochromatic photon flux, and $A(\lambda_{in})$ is the sample absorption at this wavelength. Combining eqs 2 and 3 yields

$$\eta_{ext}(\lambda_{in}) = \frac{1}{\phi_{abs}(\lambda_{in})}\int_0^d \Delta G(\lambda_{in}, z)S(z)\mathrm{d}z \tag{4}$$

Thus, combining optical modeling to calculate $\Delta G(\lambda_{in}, z)$ with the measurements of $\phi_{abs}(\lambda_{in})$ and $\eta_{ext}(\lambda_{in})$, the SELE profile can be extracted from eq 4 using regularization tools as previously applied for SCE extraction.[8,9]

Using device simulation, the SELE can be calculated directly by finding the change in the PL photon flux that is a result of a small perturbation in optical generation at a specific position within the device. For a 1D device, this follows

$$S(z) = \frac{\phi_{out}(G(z') + \Delta G\cdot\delta(z' - z)) - \phi_{out}(G(z'))}{\Delta G} \tag{5}$$

Here $\phi_{out}(G(z'))$ is the PL photon flux induced by the optical generation profile $G(z')$, and $\Delta G\cdot\delta(z' - z)$ is the additional optical generation induced at a depth of z. These simulations

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allow for studying the SELE of devices with different bulk and surface properties and correlating between various recombination mechanisms and the SELE profile features.

2. METHODS

2.1. PL and ELE Measurements. The SELE extraction is demonstrated with a GaAs wafer, which serves as a model system due to its well understood properties, strong PL signal, and simple geometry. Measurements were performed on a double-side-polished $(1.98 \pm 0.28) \times 10^{19}$ cm^{-3} zinc-doped, 350 μm thick GaAs wafer (Wafer Technology Ltd.). Incident wavelength-dependent ELE measurements were conducted in an integrating sphere-based setup, as illustrated in Figure 1a. A 75 W Xe lamp-tunable light source (Horiba PowerArc) was set to produce a monochromatic illumination with a bandwidth of 20 nm. The output beam was focused onto a 4 mm × 4 mm spot at the sample port of the integrating sphere (Labsphere RTC-060-SL). The output signal from the integrating sphere was passed through a long-pass filter (Hoya IR80) to an optical fiber (Thorlabs FG910LEC) that coupled between the integrating sphere and a spectrometer (Avantes Avaspec-ULS2048 × 64TEC-EVO). The long-pass filter was chosen to correspond to the PL range in order to exclude the incident and reflected light from the PL spectrum.

The ELE was found by integrating the PL signal over the emitted wavelengths, λ_{PL}

$$\eta_{ext}(\lambda_{in}) = \frac{1}{\phi_0(\lambda_{in})(1 - R(\lambda_{in}) - T(\lambda_{in}))} \int \Phi_{out}(\lambda_{PL}, \lambda_{in}) d\lambda_{PL} \tag{6}$$

where $\Phi_{out}(\lambda_{PL}, \lambda_{in})$ is the measured PL spectrum in response to the monochromatic illumination at wavelength λ_{in}, and $R(\lambda_{in})$ and $T(\lambda_{in})$ are the sample reflection and transmission, respectively. The incident photon flux was measured using a calibrated detector (Newport 818-UV/DB) that was placed on the sphere's sample port. The calibration of the system transfer function, which correlates between the spectrometer counts and the PL photon flux from the sample, was performed using NIST traceable calibrated diffuse reflectance standards (Labsphere). More details on the calibration method can be found in Section S1 (Supporting Information).

2.2. Optical Characterization and Modeling. Spectrophotometric reflection and transmission measurements were performed with an Agilent Cary-5000 UV−vis spectrophotometer using the UMA configuration set to an incident angle of 8°, which is the illumination angle in the ELE setup. The optical constants of the sample were obtained by using spectroscopic ellipsometry (SE) measurements (J.A. Woollam M-2000 DI) and analyzed with variable angle spectroscopic ellipsometer (VASE) software. The SE measurements were conducted at incident angles of 65, 70, and 75°, and the ellipsometer reflection intensity was calibrated against a known Si/SiO_2 standard. The samples were modeled as a stack of two layers, consisting of a GaAs substrate and a 2.7 nm native oxide surface layer.[24] Free carrier absorption (FCA) was accounted for by using the Drude model.

Although light absorption is attenuated by FCA, the generation of photons in spontaneous emission is determined by wavelength-dependent absorption coefficients that are not affected by FCA.[25] We therefore distinguish between the absorption coefficient of the bulk that relates to light emission, α_b, and the absorption coefficient, α, that includes FCA.[26]

Assuming a homogeneous sample with no absorption in the boundary layers, the measured optical constants were verified against the reflectance and transmittance measurements of the sample. Details on the verification method and ellipsometry model can be found in Section S2 (Supporting Information).

2.3. SELE Extraction. The SELE was extracted using a Tikhonov regularization process as previously used for SCE extraction.[8,9] In this process, eq 4 is first discretized

$$\bar{\eta}_{ext}(\lambda_{in}) = \frac{1}{\bar{\phi}_{abs}(\lambda_{in})} \overline{\overline{\Delta G}}(\lambda_{in}, z_i) \bar{S}(z_i) \tag{7}$$

In this case, the device thickness was divided into one-dimensional spatial elements of various widths such that $\overline{\overline{\Delta G}}(\lambda_{in}, z_i)$ is a matrix that describes the total optical generation within the element located at position z_i that is induced by monochromatic illumination with wavelength λ_{in}, and $\bar{S}(z_i)$ is a vector of the average SELE within the element z_i.

Since eq 7 has an infinite number of solutions, the candidate solutions must be constrained. The measured incident wavelength-dependent ELE, $\bar{\eta}_{ext}(\lambda_{in})$, and the modeled optical generation profile, $\overline{\overline{\Delta G}}(\lambda_{in}, z_i)$, are inserted into eq 7, and the extracted SELE vector, $\bar{S}(z_i)$, is found to minimize the following expression

$$\min\left\{ \left\| \frac{(\overline{\overline{\Delta G}}(\lambda_{in}, z_i))\bar{S}(z)}{\bar{\phi}_{abs}(\lambda_{in})} - \bar{\eta}_{ext}(\lambda_{in}) \right\|_2 + \kappa^2 \|L(\bar{S}(z))\|_2 \right\} \tag{8}$$

The first term in the parentheses defines the accuracy in which the candidate solution, $\bar{S}(z)$, reproduces the measured ELE as described with eq 7. L is an operator such that the second term in parentheses constrains the physical properties of the solution. In this work, the second derivative operator is chosen to constrain the curvature of the solution. The regularization parameter κ is a weight used to balance between a solution that reproduces the measured ELE accurately while maintaining physical features according to the chosen L operator. The candidate solutions are SELE vectors that minimize (8), each corresponding to a different value of κ. From this group of solutions, those that produce both a small deviation from measurements and are smooth, i.e., have a small second derivative, are averaged to produce the mean profile, which is the extracted SELE. More details on the regularization process and method can be found in Section S3 (Supporting Information).

2.4. ELE and SELE Simulation. SELE profiles were also computed directly from simulations using eq 5. To simulate the emission PL spectrum while accounting for charge carrier drift, diffusion, generation, and recombination processes, the semiconductor continuity equations for electrons and holes were coupled to the Poisson equation and to an optical model while considering the redistribution of photons through photon recycling. First, a finite-element partial differential equation solver (COMSOL Multiphysics) was used to calculate the potential distribution and the concentration of electrons and holes at every depth within the device. Under the assumption of isotropically radiated photons, the electron and hole concentrations were used to calculate the spontaneous emission rate per solid angle at every depth, following[25,27]

$$dR_{rad}(z, E) = \frac{1}{4\pi^3\hbar^3 c_0^2} \alpha_b(E) n(E)^2 E^2 \frac{1}{\exp\left(\frac{E - (E_{F_C}(z) - E_{F_V}(z))}{k_B T}\right) - 1} \tag{9}$$

where n is the sample's refractive index, and E_{F_C} and E_{F_V} are the electron and hole quasi-Fermi levels. α_b (E) is the energy-dependent absorption coefficient of the bulk that does not consider FCA (see Sections 2.2 and S2 in the Supporting Information). The spatial distribution of the quasi-Fermi levels is found by solving the continuity equations, thus accounting for the effects of recombination and transport mechanisms.

Radiated photons are either reabsorbed, which contributes to the photogeneration, or emitted through the front or back surfaces, thus contributing to the PL signal. The contribution of reabsorbed photons to the optical generation is calculated in an iterative process that produces the spontaneous emission rate, $dR'_{rad}(z, E)$, which accounts for photon recycling within the material. The PL spectrum from the front or back surface of the device is then calculated using the

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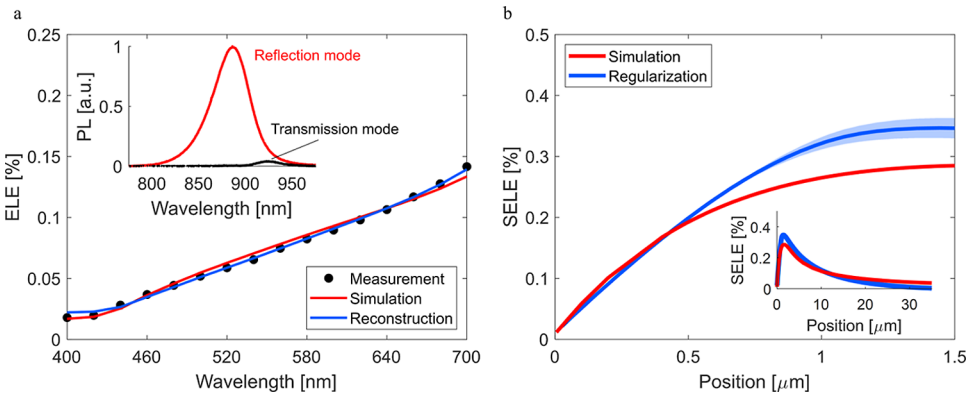

Figure 2. ELE and SELE profiles of a p-type GaAs wafer. (a) Incident wavelength-dependent ELE measurement, simulation, and reconstruction from the extracted SELE profile. ELE measurements were taken in reflection mode. The inset shows a comparison of reflection and transmission mode PL measurements for an incident wavelength of 600 nm. The simulation parameters were set to fit best the ELE measurement. (b) Simulated and extracted SELE profiles. The shaded area is the standard deviation of all of the candidate solutions.

corrected spontaneous emission rate multiplied by photon escape probabilities, following

$$\Phi_{\mathrm{out}_{\mathrm{sim}}}^{\mathrm{front/back}}(E) = \int_0^d \int_0^{2\pi} \int_0^{\pi} \mathrm{d}R'_{\mathrm{rad}}(z, E) P_{\mathrm{esc}}^{\mathrm{front/back}}(z, E, \theta, \varphi) \sin(\theta)\mathrm{d}\theta\, \mathrm{d}\varphi\, \mathrm{d}z \quad (10)$$

where $P_{\mathrm{esc}}^{\mathrm{front/back}}(z, E, \theta, \varphi)$ is the probability for a photon to escape from the front or back surface. E is the photon energy, z is the depth from the surface in which the photon was generated, θ is the photon propagation angle with respect to the surface normal, and φ is the azimuthal angle. Details on the photon recycling calculation model and PL simulations can be found in Section S4 (Supporting Information). Once the photon flux is calculated, the incident wavelength-dependent ELE, $\eta_{\mathrm{ext}}^{\mathrm{sim}}(\lambda_{\mathrm{in}})$, and the SELE can then be calculated using eqs 3 and 5 respectively. SELE simulation details can be found in Section S5 (Supporting Information).

3. RESULTS AND DISCUSSION

The SELE of GaAs wafers was extracted in three steps: first, an optical model was constructed to calculate the wavelength-dependent optical generation profile $\Delta G(\lambda_{\mathrm{in}}, z)$; next, the wavelength-dependent ELE was obtained from PL measurements; and finally, the SELE was extracted according to eq 4. The optical constants of the samples were obtained with spec-ellipsometry analysis as described in Section 2, and the optical generation profile was calculated assuming ray optics. The incident wavelength-dependent PL spectra were measured in an integrating sphere system using a spectrometer and illuminated with a xenon lamp coupled to a monochromator. Figure 1a shows an illustration of the experimental setup, and Figure 1b shows the measured incident wavelength-dependent PL spectra. More details on the setup and its calibration can be found in Section 2.1. Figure 2a shows the incident wavelength-dependent ELE spectrum calculated by using the measured PL spectra (Figure 1b) and eq 3. The SELE profile of the GaAs wafer was extracted from this ELE spectrum and the optical model using Tikhonov regularization, as described in Section 2. Figure 2b shows the extracted SELE profile near the surface, and the inset shows the profile deeper within the wafer. The solid curve represents the mean profile that is obtained from all the solutions that are both physically constrained and can reconstruct the measurements. The shaded area is the standard deviation of all these solutions.

The SELE profile maps the competition between radiative and nonradiative recombination processes as well as the probability for photons that are generated within the bulk of the device to be emitted. Near the surface, the low SELE is a result of surface recombination through defects suppressing radiative recombination processes. As a result, the SELE initially increases with the distance from the surface. However, deeper within the material, radiative recombination photons are less likely to be emitted due to self-absorption, and the SELE decreases. A considerable portion of the photoluminescence photons is at energies slightly below the band gap. Since these photons are hardly reabsorbed even when generated far from the surface, the decay of the SELE with the distance from the surface is significantly more moderate than that dictated by the absorption coefficient for photons with energy higher than the band gap. The nonzero SELE deep within the sample was validated experimentally with transmission mode PL measurements. In these experiments, the input beam is incident on one surface of the sample, but only photons emitted off the opposite surface are collected by the integrating sphere. The inset in Figure 2a shows the PL spectra measured in reflection mode and transmission mode for an incident wavelength of 600 nm. Both spectra were normalized to the maximum value of the reflection mode PL spectrum. The nonzero PL measured in transmission mode implies that optical generation near one surface leads to photon emission off the other. Thus, the SELE is higher than zero even near the back surface of the sample and consequently all through the device.

The extracted SELE profile was validated by reconstructing the incident wavelength-dependent ELE spectrum with finite-element simulation. The reconstructed ELE spectrum is calculated by inserting the extracted SELE and the optical generation profiles into eq 4. The resulting ELE spectrum is shown in Figure 2a. Since optical modeling errors and inadequate regularization procedures result in deviations between the measured and reconstructed spectra,[8,9] the excellent agreement between the spectra supports the validity of the regularization process and the extracted SELE profile.

The extracted SELE profile was further validated with device simulations in which the emitted PL spectrum could be calculated for any optical generation profile and material properties. A detailed description of the device simulation can be found in Section 2.4. The incident wavelength-dependent ELE spectrum was calculated by providing the simulation with optical generation profiles corresponding to monochromatic

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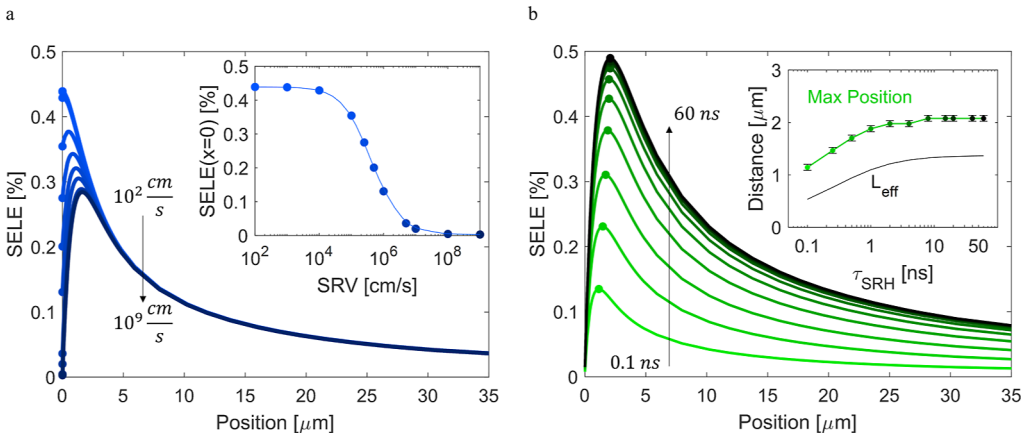

Figure 3. Simulated SELE profiles of p-type GaAs wafers. (a) SELE profiles for wafers with surface recombination velocity between 10^2 and 10^9 cm/s and τ_{SRH} of 0.4 ns. The inset shows the SELE at the surface as a function of the surface recombination velocity. The color coding of the markers matches that of the SELE profiles. (b) Simulated SELE profiles for wafers with τ_{SRH} between 0.1 and 60 ns and a surface recombination velocity of 2×10^7 cm/s. The inset shows the peak position of each curve and the calculated effective diffusion length. The error bars correspond to the width of the simulated element at the maximum position. All of the other input parameters used for the simulation are detailed in Section S5 (Supporting Information).

illumination inputs, and the SELE is found using delta-function-like optical generation profiles according to eq 5. The sample material properties were fitted to best match the emitted PL spectrum and the incident wavelength-dependent ELE. The simulated ELE is shown in Figure 2a, and a comparison between the simulated and measured PL spectra can be found in Section S2 (Supporting Information). The material properties used in this simulation are listed in Table S5.1 (Supporting Information). Figure 2b shows the simulated SELE profile. Within the first 500 nm from the surface, where the vast majority of the charge carriers are generated, there is excellent agreement between the extracted SELE and the SELE profile obtained from the simulation. The model underestimates the SELE deeper within the device and as a result predicts a slightly lower ELE at longer incident wavelengths (Figure 2a).

To correlate between different features in the SELE profile and material properties, SELE profiles were simulated for GaAs wafers with similar optical properties as the measured sample but with various surface recombination velocities and Shockley Read Hall (SRH) lifetimes (all the other material properties are as in Table S5.1). Figure 3a shows the simulated SELE profiles with different surface recombination velocities. The minority carrier bulk lifetime is 0.4 ns. As expected, higher surface recombination reduces the SELE at the surface, while the SELE within the device is unaffected by the change in the surface recombination velocity. Under the extreme case of long diffusion length and very high surface recombination velocity, the transport of charge carriers that are generated as deep as 5 μm from the surface is still affected by surface recombination. The inset shows the SELE at the surface as a function of the surface recombination velocity. The SELE at the surface is extremely sensitive to the surface recombination for surface recombination velocities above 10^4 cm/s. Thus, in this range, the effect of surface treatments can be quantified by analyzing the SELE at the surface. For surface recombination velocities below 10^4 cm/s, surface recombination is not the dominant recombination process, and the SELE at the surface is determined by other mechanisms. Figure 3b shows the computed SELE profiles for various minority carrier SRH lifetimes and a surface recombination velocity of 2×10^7 cm/s. Increasing the minority carrier lifetime increases the SELE in the bulk of the device because of an increase in the internal luminescence efficiency. However, due to the high surface recombination velocity, the SELE at the surface remains almost unchanged. The drop in the SELE toward the surface and the position of the SELE peak are a result of surface recombination of charge carriers that are generated within the bulk of the material and diffuse toward the surface where they recombine. Thus, when surface recombination is the dominant loss mechanism, the position of the maximum value of SELE is comparable to the diffusion length. The inset in Figure 3b shows the effective diffusion length, $L_{eff} = \sqrt{D\tau_{eff}}$, and the maximum SELE position as a function of the SRH lifetime. Here, the effective lifetime is assumed to follow $\tau_{eff}^{-1} = \tau_{SRH}^{-1} + \tau_{Rad}^{-1} + \tau_{Aug}^{-1}$, where τ_{Rad} is the radiative lifetime and τ_{Aug} is the Auger lifetime. Both lengths show a very similar trend saturating for high values of τ_{SRH}, where the radiative lifetime becomes the dominant factor determining the effective minority carrier lifetime and diffusion length. This exemplifies how the analysis of the SELE profile can shed light on charge transport mechanisms within the sample.

The buildup of the obtainable photovoltage within a device is determined by the contribution of different regions to the external luminescence. According to eq 1, the deviation from an ideal open-circuit voltage follows $\Delta V_{oc} = V_{oc} - V_{oc,ideal} = kT/q \ln(\eta_{ext})$. Inserting eq 4 into the expression for ΔV_{oc} yields

$$\Delta V_{oc}(d) = \frac{kT}{q} \ln\left(\int_0^d \frac{\Delta G(z) S(z)}{\phi_{abs}} \, dz \right) \tag{11}$$

The term in the integral can be described as the spatial contribution to the photovoltage, and the buildup of the photovoltage within the device can be found by integrating this term over various device thicknesses. The spatial contribution to the photovoltage that can be obtained with an absorber layer that is made of the tested GaAs was estimated using the extracted SELE profile from Figure 2b and an optical generation profile corresponding to AM1.5 illumination.[28] The obtainable photovoltage buildup, $\Delta V_{oc}(d)$, was then calculated using eq 11 for various absorber layer thicknesses. Figure 4 shows the spatial contribution to the photovoltage

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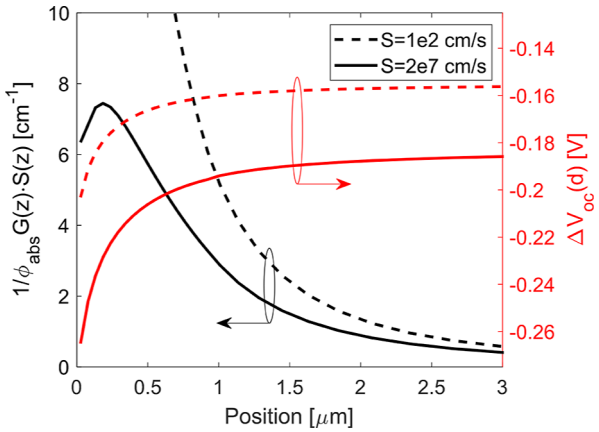

Figure 4. Spatial contribution to ΔV_{oc} (left vertical axis), and ΔV_{oc} as a function of the device thickness (right vertical axis) when the surface recombination is significant ($S = 2 \times 10^7$ cm/s, solid curves) and when the surface recombination is negligible ($S = 100$ cm/s, dashed curves). The optical generation profile corresponds to AM 1.5 illumination, and the SELE profile is as in Figures 2b and 3a.

(left vertical axis) and $\Delta V_{oc}(d)$ for different absorber layer thicknesses (right vertical axis). Almost the entire photovoltage is built within 2 μm next to the surface, providing an upper bound for the thickness of an absorber layer made of this material. Also shown in Figure 4 are the photovoltage contribution and ΔV_{oc} calculated for a material with a negligible surface recombination ($S = 100$ cm/s). The difference in ΔV_{oc} between the two curves allows the estimation of the voltage losses due to surface recombination. For example, for a thickness of 0.5 μm, the ΔV_{oc} difference is 39.8 mV and for a thickness of 2 μm, the ΔV_{oc} difference is 30.5 mV.

Since the photovoltage buildup with depth is estimated by integrating the same function over different thicknesses, it does not account for changes in the SELE due to recombination at the assumed back surface. As shown in Figure 3a, a high surface recombination velocity can reduce the SELE as far as 5 μm from the surface. Hence, the $\Delta V_{oc}(d)$ buildup in Figure 4 serves as an upper bound for the photovoltage assuming an ideal back surface.

4. CONCLUSIONS

Analysis of the SELE can provide a wealth of information on loss mechanisms, self-absorption processes, and the contribution of different regions to the photovoltage that can be obtained from a sample. We have shown how the SELE profile can be extracted by combining incident wavelength-dependent PL measurements with optical modeling and demonstrated the extraction method on a GaAs wafer. The extraction method was verified with device simulations. The SELE near the surface is highly sensitive to surface recombination, and further away from the surface, it is determined by the internal luminescence efficiency and the self-absorption of photons. Since the SELE analysis can quantify the performance of surface passivation layers as well as the spatial contribution to the photovoltage, it directly complements other operando methods, such as the spatial collection efficiency analysis that maps charge transport mechanisms. Thus, SELE analysis can provide a key step toward rapid optimization of materials and interfaces for photovoltaic cells.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsaem.4c00355.

Additional information regarding the photoluminescence calibration method, validation of the optical constants, regularization method, photon recycling model, and SELE simulations (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Gideon Segev − *School of Electrical Engineering, Tel-Aviv University, Tel-Aviv 6997801, Israel*; orcid.org/0000-0002-6175-3458; Email: gideons1@tauex.tau.ac.il

Authors

Tamir Yeshurun − *School of Electrical Engineering, Tel-Aviv University, Tel-Aviv 6997801, Israel*

Mor Fiegenbaum-Raz − *School of Electrical Engineering, Tel-Aviv University, Tel-Aviv 6997801, Israel*

Complete contact information is available at: https://pubs.acs.org/10.1021/acsaem.4c00355

Notes

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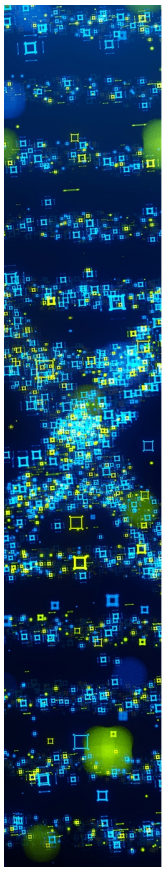
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